FTM Multilateration Analysis

Comprehensive Performance Report

WiFi Fine Timing Measurement (802.11mc)

Indoor Positioning System Evaluation

```
Test Configuration:
• Room: 20m × 10m
• Responders: 4 ESP32-S3 devices at corners
• Positions tested: 8 locations
• Samples per position: ~1000 cycles
• Frame count: 16, Burst period: 4
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Generated: March 05, 2026

Executive Summary

Key Findings

This report analyzes Fine Timing Measurement (FTM) performance for indoor positioning using 4 ESP32-S3 responders in a 20m × 10m room.

Main Results (All 4 Responders)

- Mean distance estimation error: +503 cm (systematic overestimation)
- Mean absolute distance error: 523 cm
- Mean position error (multilateration): 5.17 m

Results Without R4 (Problematic Responder Excluded)

- Mean distance estimation error: +343 cm
- Mean absolute distance error: 362 cm (+30.7% improvement)
- Mean position error (trilateration): 4.62 m (+10.8% improvement)

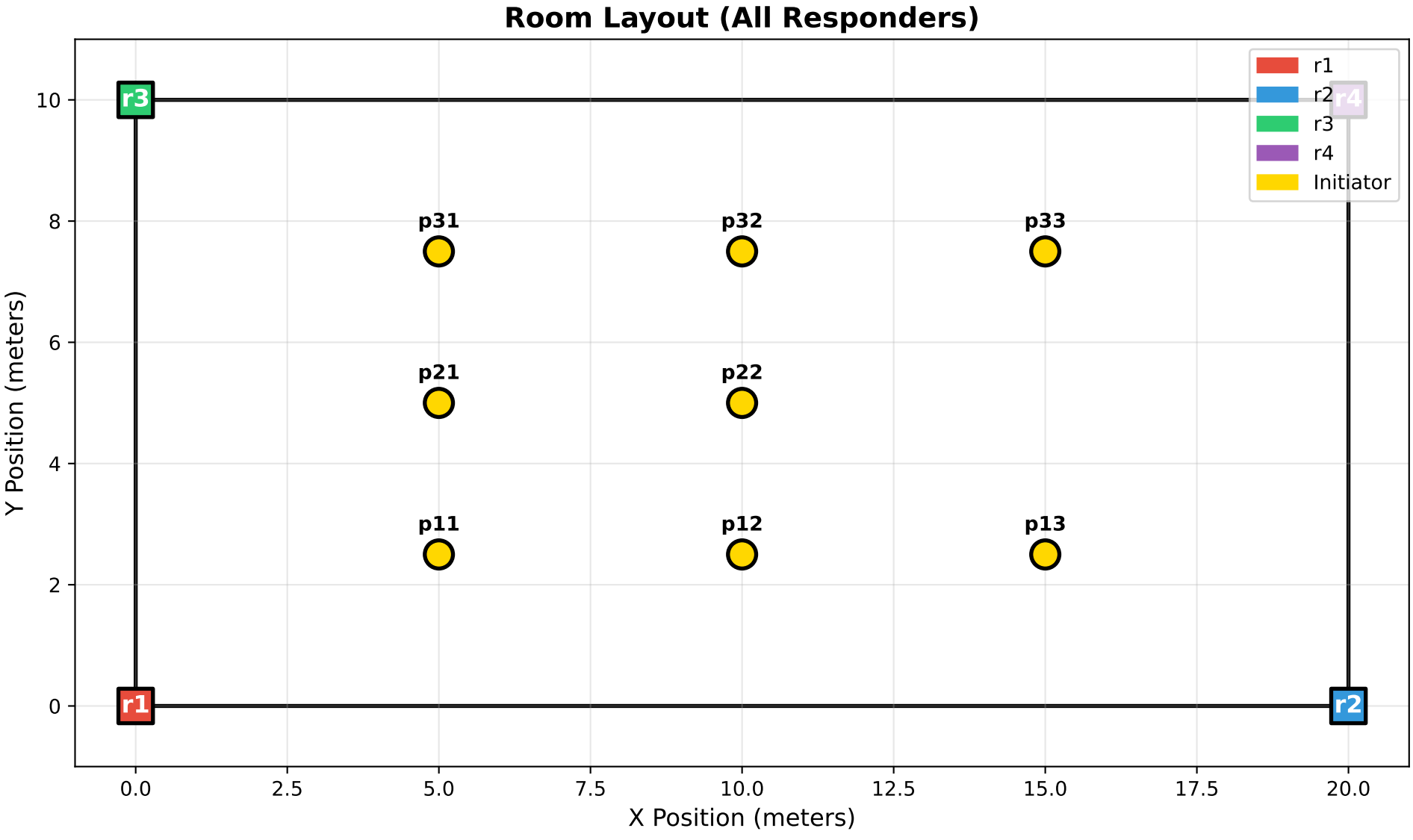
Critical Observations

- R4 had 3.6× worse accuracy than best responder (r3)
- External antenna (r3) performs ~2× better than PCB antennas
- Position near windows/doors (r2) shows increased error
- System reliability: 94.8% complete samples (all 4 responders)

Hardware Configuration

Responder hardware and measured performance metrics

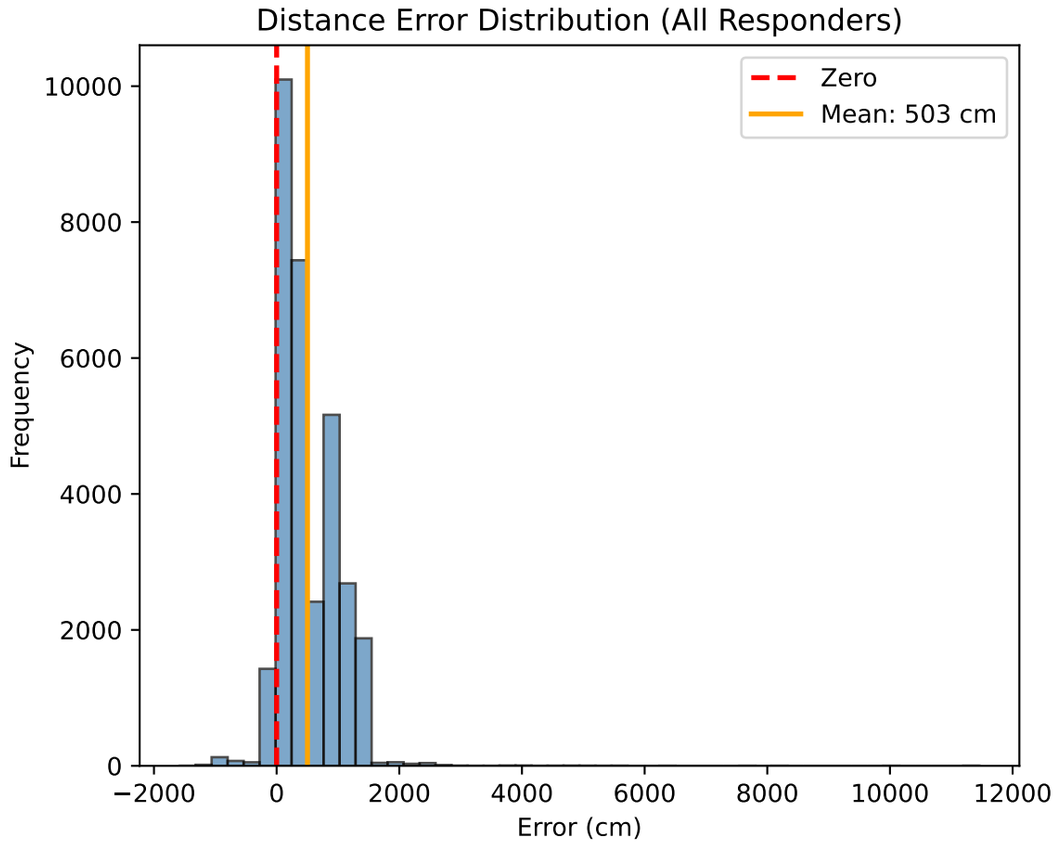
ID	Hardware	Antenna	Position	Mean |Error|	Mean RSSI
r1	Adafruit Feather ESP32-S	PCB	(0, 0)	277 cm	-41.4 dBm
r2	Adafruit Feather ESP32-S	PCB	(20, 0)	527 cm	-45.0 dBm
r3	Seeed XIAO ESP32-S	u.FL External	(0, 10)	283 cm	-34.7 dBm
r4	Adafruit Feather ESP32	PCB	(20, 10)	1005 cm	-49.7 dBm

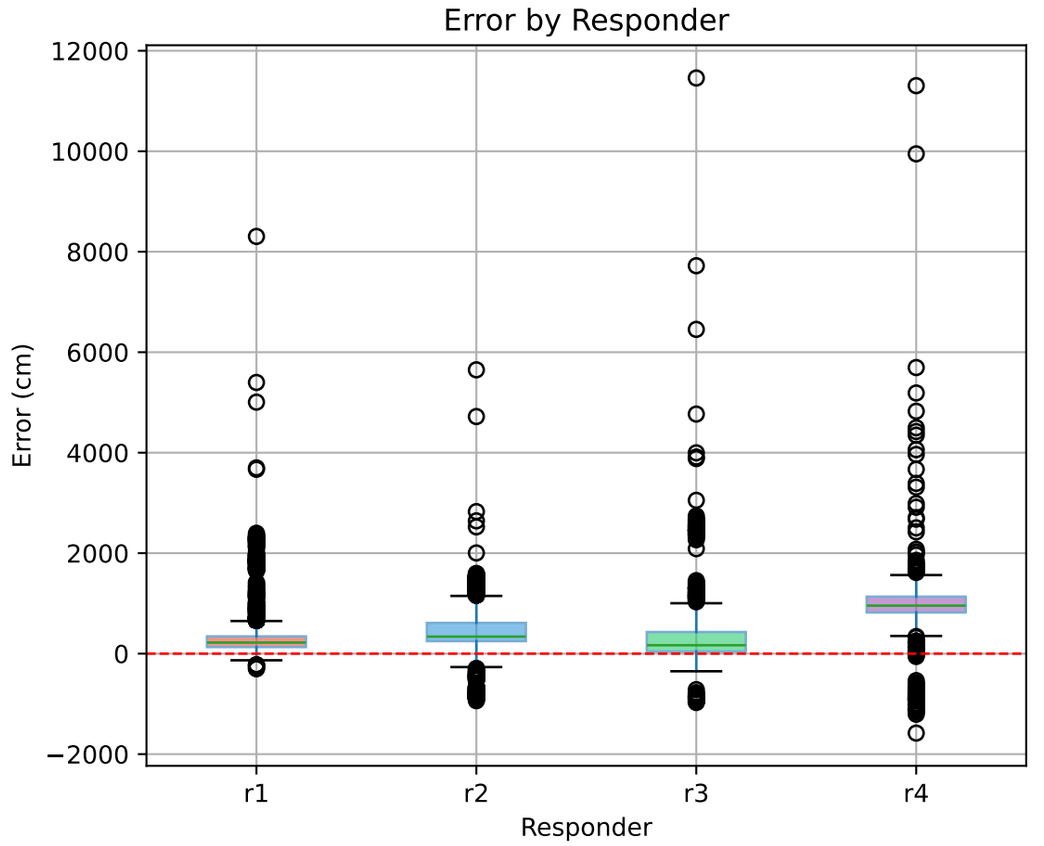

Room Layout (All Responders)
r1
r2
r3
r4
Initiator
p31
p32
p33
p21
p22
p11
p12
p13
Y Position (meters)
X Position (meters)
0
2
4
6
8
10
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

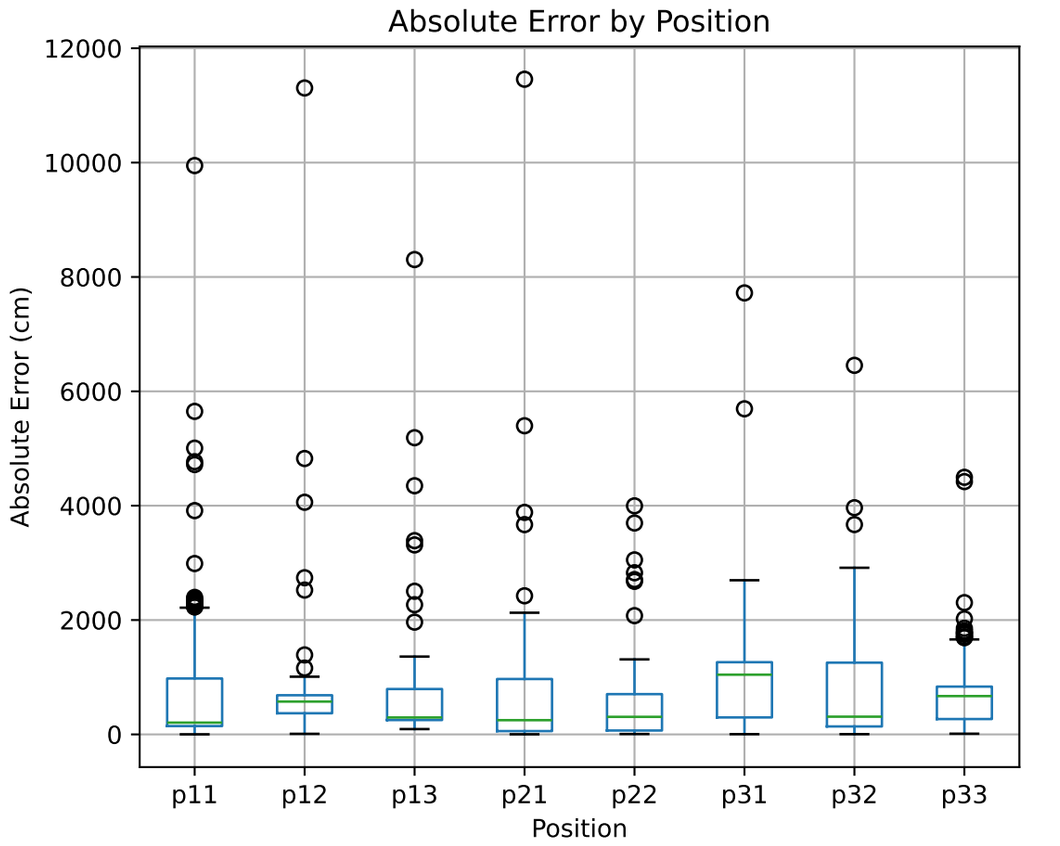
Distance Estimation Error by Responder

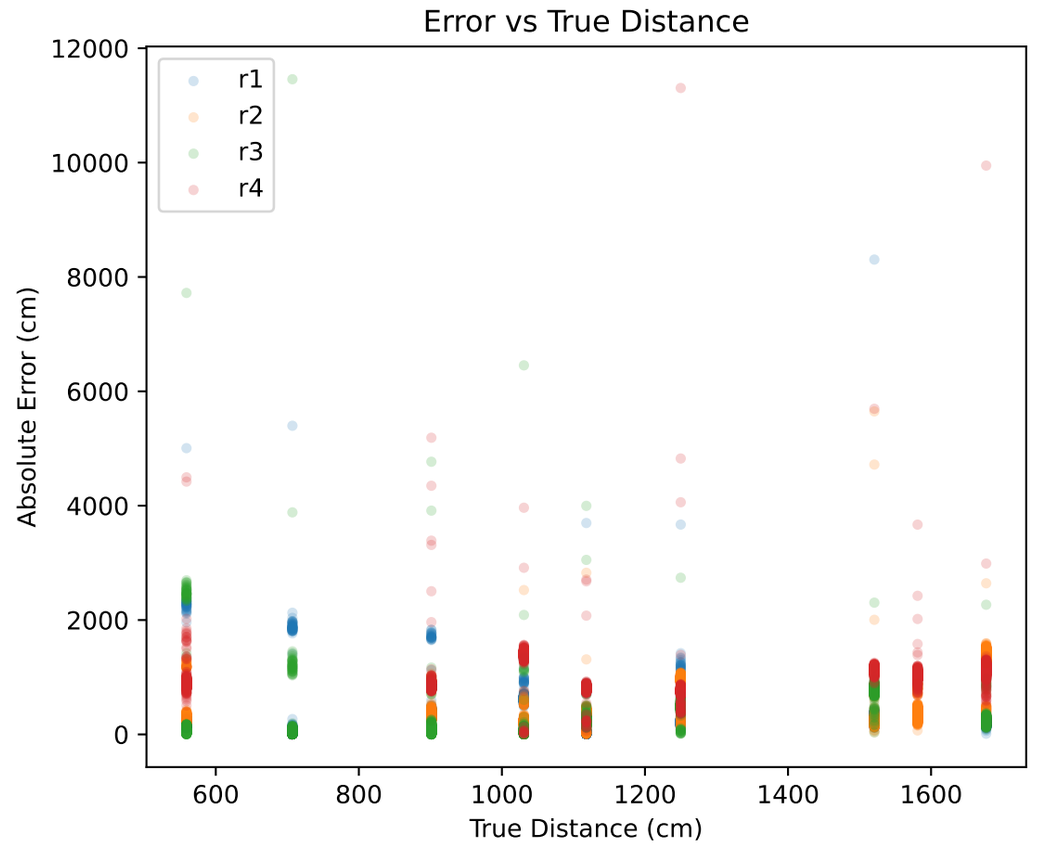
With all 4 responders

Responder	Mean Error (cm)	Std Dev (cm)	Mean |Error| (cm)	Mean RSSI (dBm)
r1	+273	304	277	-41.4
r2	+499	445	527	-45.0
r3	+256	385	283	-34.7
r4	+985	361	1005	-49.7


Distance Error Distribution (All Responders)
Zero
Mean: 503 cm
Frequency
Error (cm)
10000
8000
6000
4000
2000
0
−2000
0
2000
4000
6000
8000
10000
12000


Error by Responder
Error (cm)
Responder
r1
r2
r3
r4
12000
10000
8000
6000
4000
2000
0
−2000

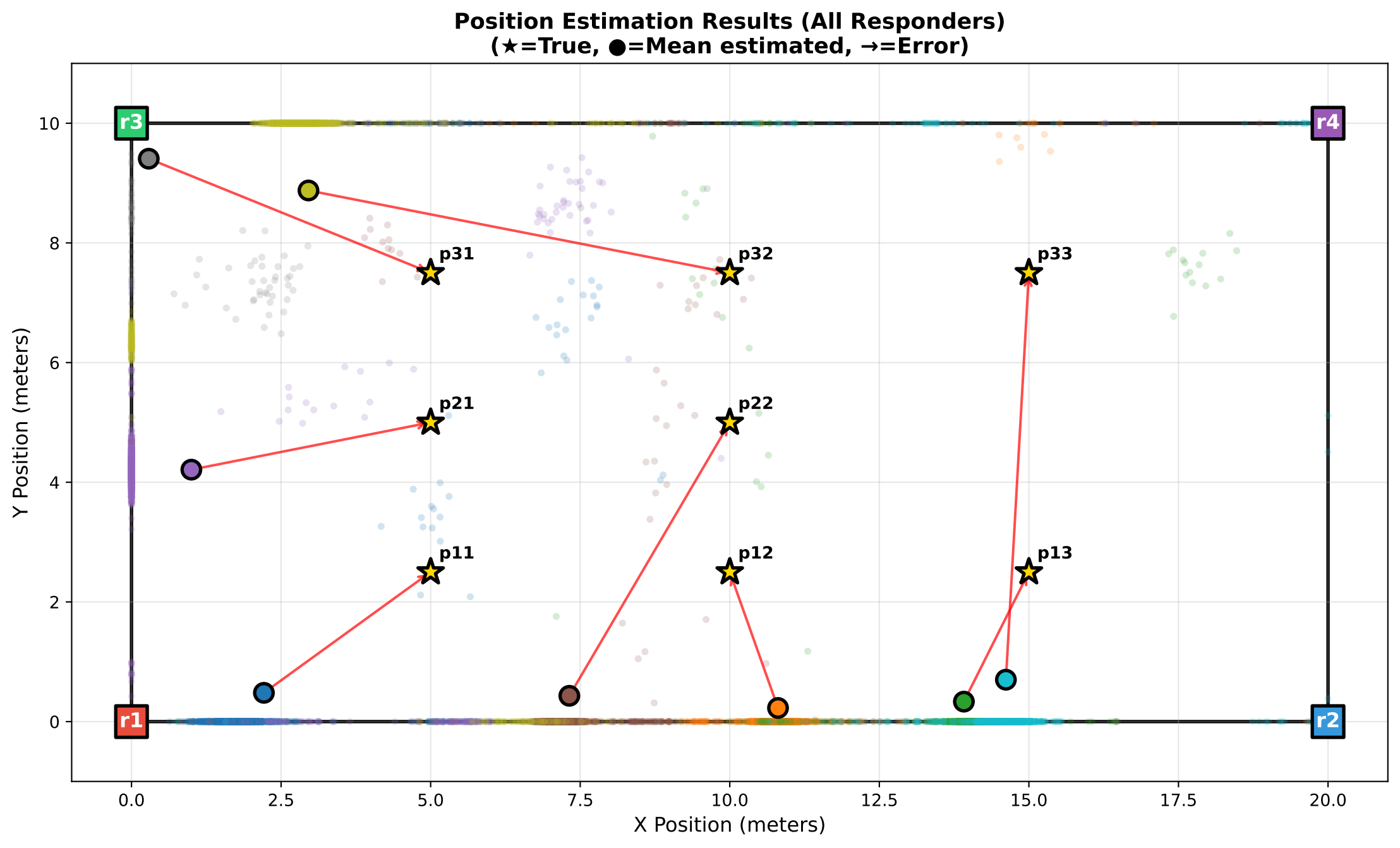

Absolute Error by Position
Absolute Error (cm)
Position
p11
p12
p13
p21
p22
p31
p32
p33
12000
10000
8000
6000
4000
2000
0

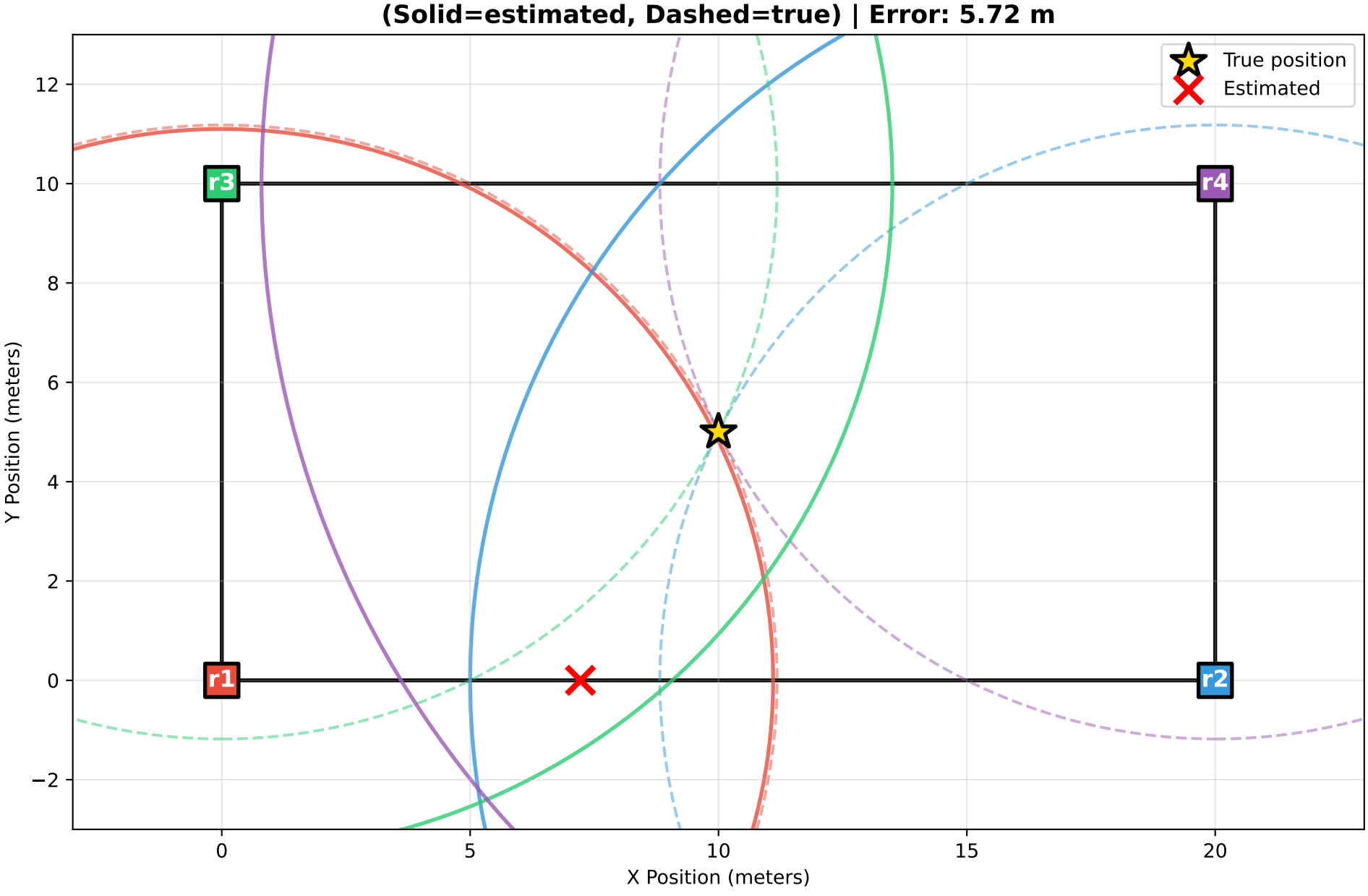

Error vs True Distance
r1
r2
r3
r4
Absolute Error (cm)
True Distance (cm)
600
800
1000
1200
1400
1600
12000
10000
8000
6000
4000
2000
0

Position Estimation Error by Location

Multilateration with 4 responders

position	Mean (m)	Std (m)	Min (m)	Max (m)	X	Y
p11	4.142	1.017	0.42	15.207	5.0	2.5
p12	2.842	1.08	2.5	10.321	10.0	2.5
p13	2.983	1.238	2.5	15.207	15.0	2.5
p21	5.098	0.729	0.935	15.811	5.0	5.0
p22	5.651	0.762	0.597	11.18	10.0	5.0
p31	5.528	1.08	2.1	16.771	5.0	7.5
p32	7.813	1.239	2.508	12.5	10.0	7.5
p33	7.335	0.981	2.505	16.112	15.0	7.5


Position Estimation Results (All Responders)
(★=True, ●=Mean estimated, →=Error)
r1
r2
r3
r4
p11
p12
p13
p21
p22
p31
p32
p33
Y Position (meters)
X Position (meters)
0
2
4
6
8
10
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0


Distance Rings: p22, Sample 1
(Solid=estimated, Dashed=true) | Error: 5.72 m
True position
Estimated
r1
r2
r3
r4
Y Position (meters)
X Position (meters)
12
10
8
6
4
2
0
−2
0
5
10
15
20

Analysis Without R4

Rationale for Excluding R4

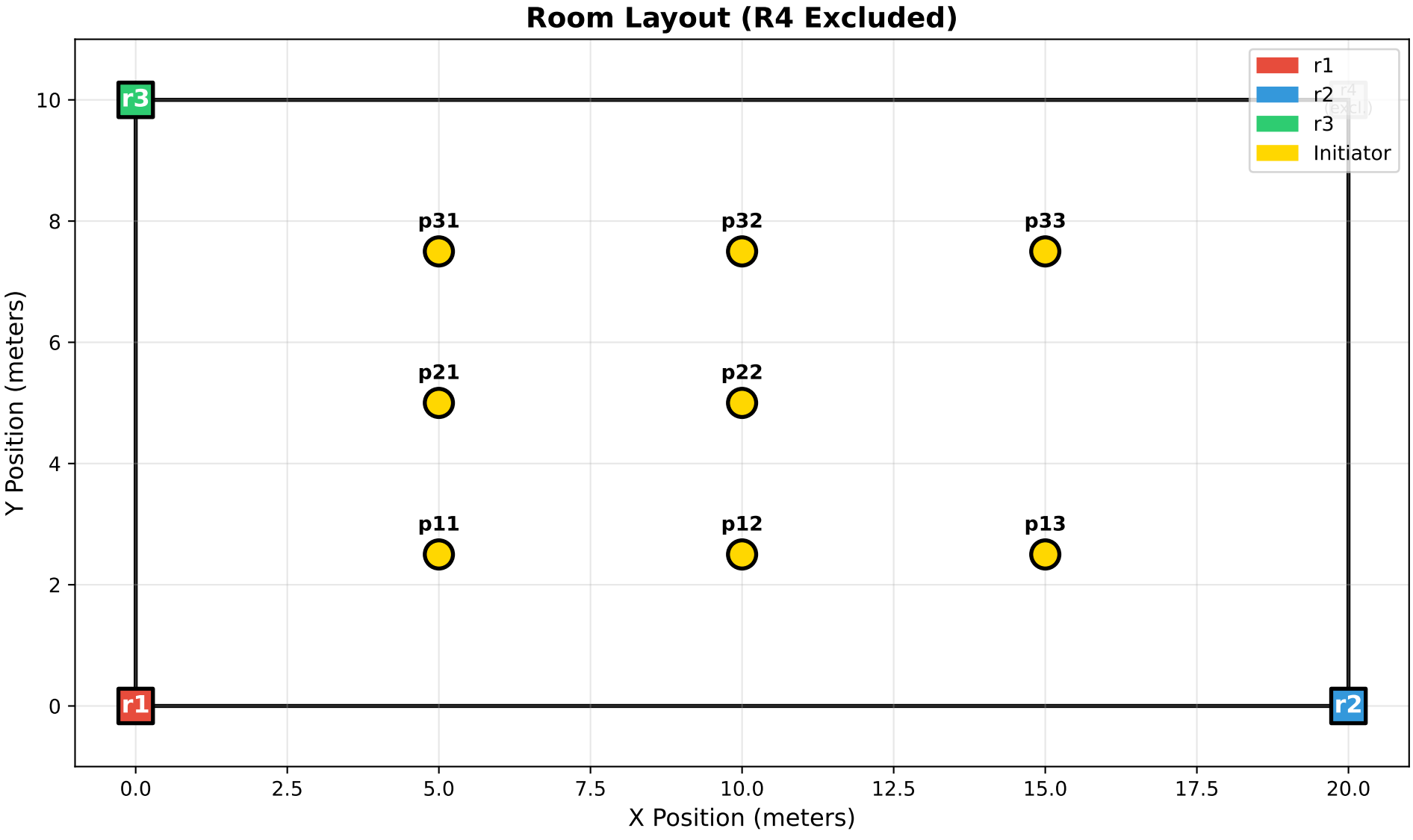
Responder R4 showed significantly degraded performance:

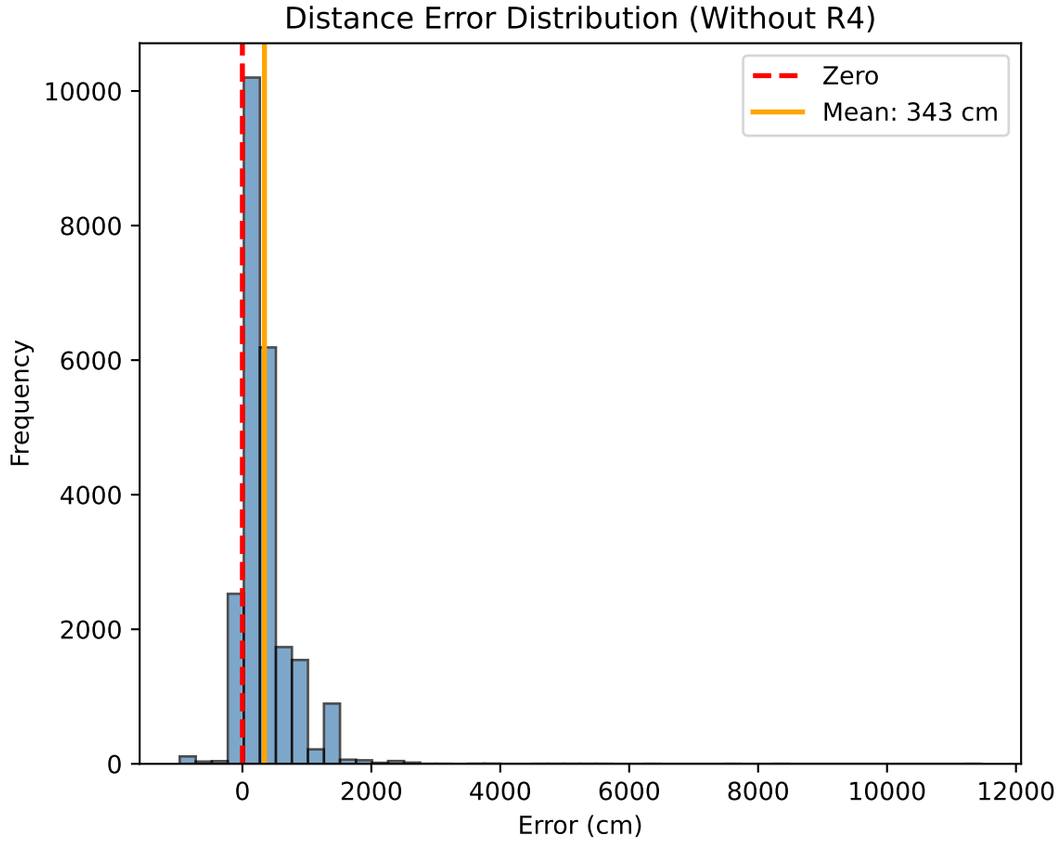
- Mean absolute error: 1005 cm (vs 280 cm for r3)
- Weakest RSSI: -49.7 dBm (vs -34.7 dBm for r3)
- Highest failure rate: 1.50%

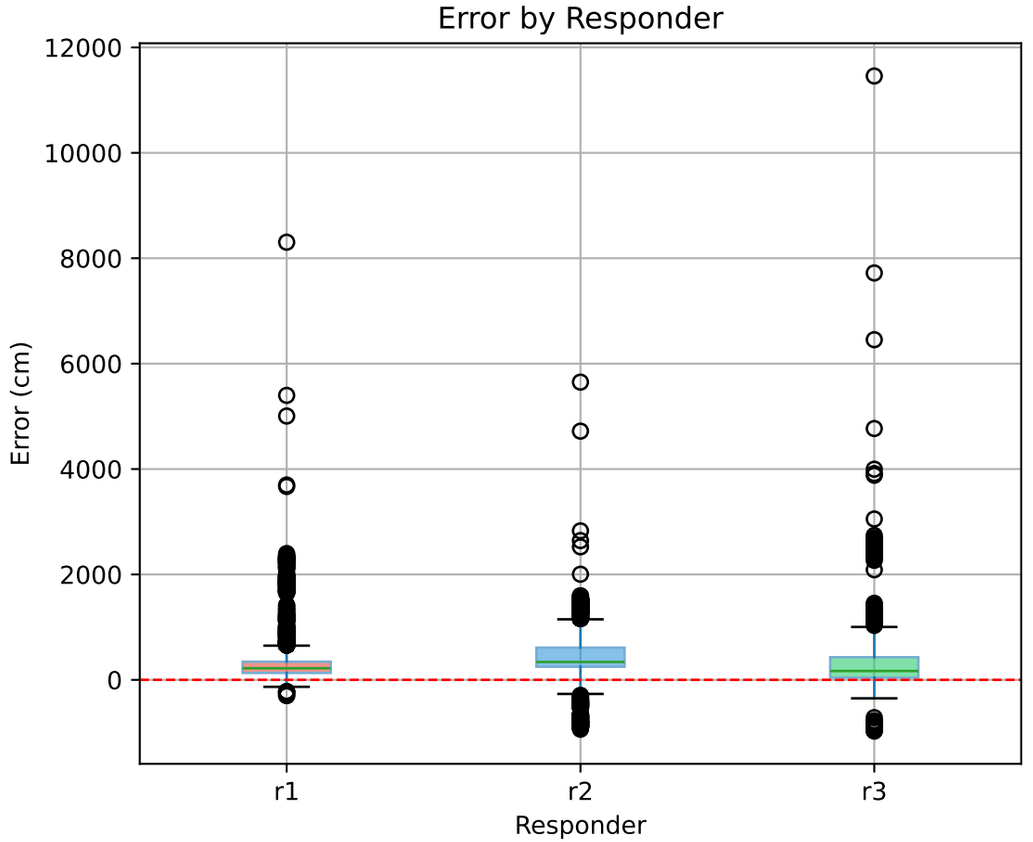
Possible causes: PCB antenna issues, environmental interference, or hardware defects. This section analyzes system performance with R4 excluded to assess achievable accuracy.

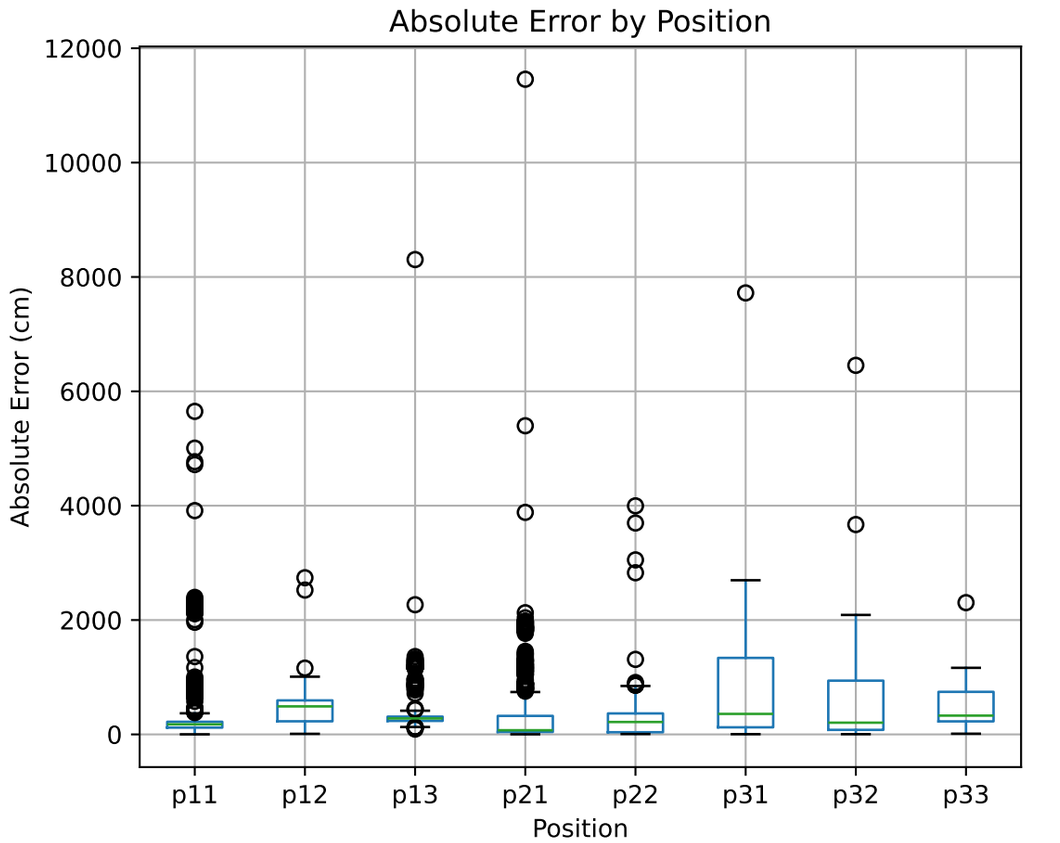
Results Summary (Without R4)

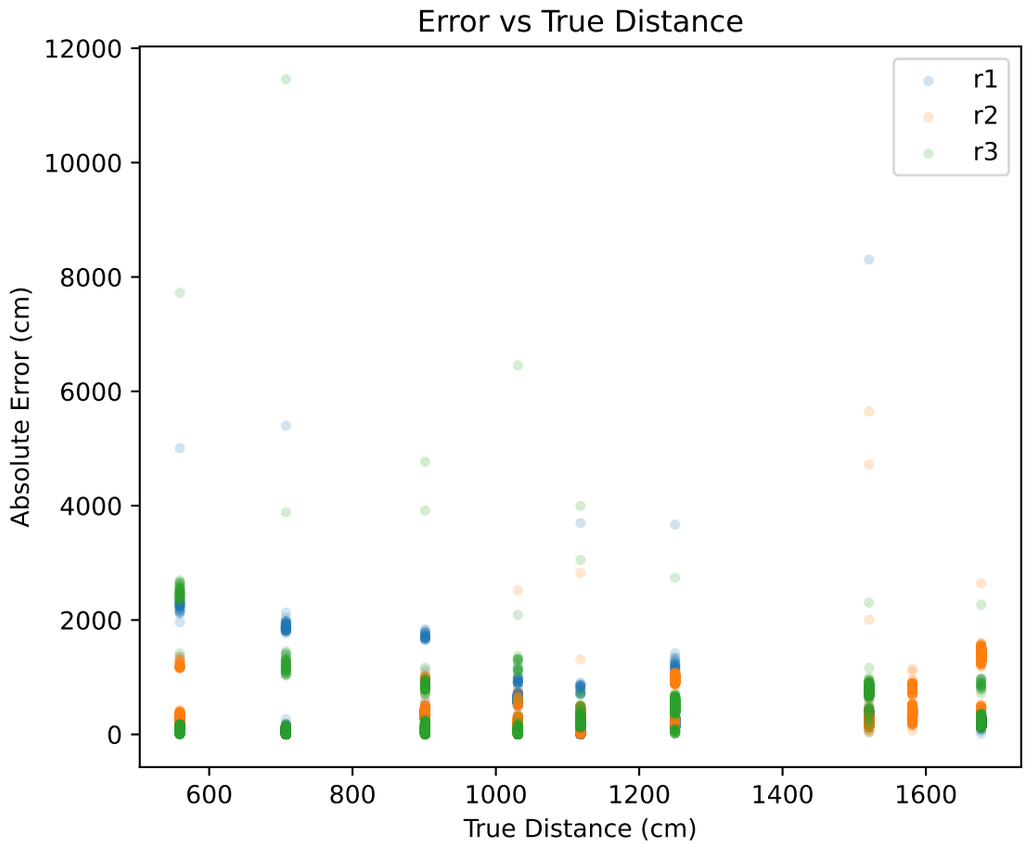
- Distance measurements: 23,696 (R4 removed: 7,880)
- Mean distance error: +343 cm
- Mean absolute error: 362 cm
- Position estimates: 7,698
- Mean position error: 4.62 m

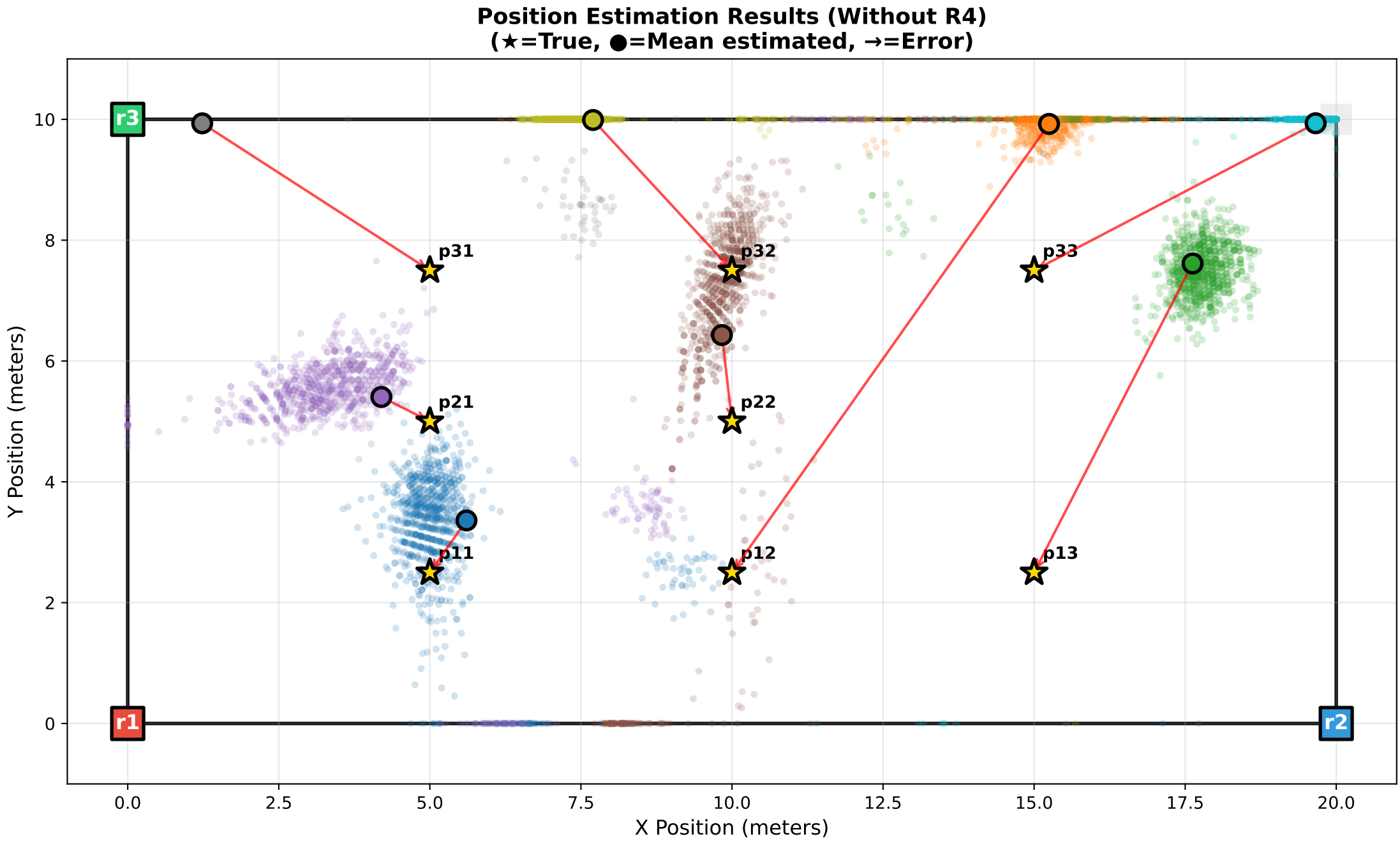

Room Layout (R4 Excluded)
r1
r2
r3
Initiator
r3
r1
r2
p31
p32
p33
p21
p22
p11
p12
p13
Y Position (meters)
X Position (meters)
0
2
4
6
8
10
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0


Distance Error Distribution (Without R4)
Zero
Mean: 343 cm
Frequency
Error (cm)
0
2000
4000
6000
8000
10000
12000


Error by Responder
Error (cm)
Responder
r1
r2
r3
0
2000
4000
6000
8000
10000
12000


Absolute Error by Position
Absolute Error (cm)
Position
p11
p12
p13
p21
p22
p31
p32
p33
0
2000
4000
6000
8000
10000
12000


Error vs True Distance
r1
r2
r3
Absolute Error (cm)
True Distance (cm)
0
2000
4000
6000
8000
10000
12000
600
800
1000
1200
1400
1600


Position Estimation Results (Without R4)
(★=True, ●=Mean estimated, →=Error)
r3
r1
r2
p31
p32
p33
p21
p22
p11
p12
p13
Y Position (meters)
X Position (meters)
0
2
4
6
8
10
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

Performance Comparison: With vs Without R4

Quantitative improvement from excluding problematic responder

Metric	With R4	Without R4	Improvement
Mean Distance Error	+503 cm	+343 cm	—
Mean |Distance Error|	523 cm	362 cm	+30.7%
Std Dev (Distance)	478 cm	398 cm	+16.8%
Mean Position Error	5.17 m	4.62 m	+10.8%
Median Position Error	5.18 m	5.10 m	—
Best Position Error	0.42 m	0.06 m	—

Conclusions & Recommendations

Distance Estimation

- Systematic positive bias observed (+343 to +503 cm depending on config)
- Antenna quality significantly impacts accuracy
- External u.FL antenna (r3) achieves ~280 cm mean error
- PCB antennas achieve ~280-530 cm depending on position

Position Estimation

- Multilateration (4 resp): 5.17 m mean error
- Trilateration (3 resp, no R4): 4.62 m mean error
- Removing faulty hardware improves overall accuracy
- Best case achievable: ~6 cm position error

Recommendations

- Use external antennas for all responders
- Avoid placing responders near windows/doors
- Implement RSSI-based quality filtering
- Consider calibration to remove systematic bias
- Minimum 4 responders recommended for redundancy

Sample Completeness

- 94.8% of samples had all 4 responders
- 5.2% had 3 responders (1 failure)
- Per-responder failure rate: 1.0-1.5%

Methodology & Code Reference

Data Processing

CSV files contain FTM measurements with columns:
position, real_x, real_y, responder_ssid, responder_label,
responder_x, responder_y, dist_est_cm, rtt_est_ns, rssi, sample_num

Error lines (starting with # E W) are filtered during loading.

Key Calculations

```
# True distance calculation
true_dist = sqrt((real_x - resp_x)² + (real_y - resp_y)²)

# Error calculation
error = dist_est_cm - true_dist_cm

# Multilateration (least squares)
def residuals(point):
    x, y = point
    calc_dists = sqrt((resp_x - x)² + (resp_y - y)²)
    return calc_dists - measured_dists

result = least_squares(residuals, initial_guess)
```